

The Components of Trust for Collaborating With AI Colleagues

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ABSTRACT

AI technologies are capable of improving the performance and productivity of teams in a variety of work contexts. These advantages may be optimized when the AI agent is considered a full team member. A vital component of the agent's acceptance as a team member or colleague is the degree to which its human coworkers feel they can trust it. To explore what factors affect the perceptions of an AI agent as a trustworthy team member and a legitimate colleague, we interviewed twenty-two professionals representing various work roles. Our results revealed that the following qualities contribute to professionals' trust in AI as a colleague: a visual presence reflective of coworkers, engagement in feedback loop and team processes through human communication, and the ability for self-development. These findings contribute to the CSCW community by advancing the current understanding of human-AI teaming and informing the design of trustworthy AI agents into the workplace.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

human-AI teaming, trust, autonomous teammates

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1 INTRODUCTION

Advances in AI technology are driving various organizations to adopt and increase the use of AI systems in the workplace to reduce human workload and increase work efficiency and accuracy [10]. Humans and AI systems each provide unique skills that, when combined, can produce unprecedented results in the workplace [6]. When humans and AI work together interdependently as a team, they can significantly enhance an organization's performance [2]. However, this vision poses two challenges regarding human trust: trust in the AI system and technology, and trust of the AI as a teammate and colleague.

Despite the demonstrated benefits that AI systems can provide in complex work environments, many of these systems are underutilized or unused due to lack of trust in the technology [25] [17]. Research shows that a user's trust in an AI agent's ability to complete its specified tasks is a driving force in whether or not the technology is accepted, adopted, and utilized [1]. This concern is always present in human interaction with new technologies, but is exacerbated in work environments [20].

A growing body of research has shown that the effectiveness of a human-AI teaming is impacted by whether the AI is viewed as a legitimate team member or merely a tool [26]. This suggests that human factors, such as trust of a colleague in organizational settings [14], may extend to human perceptions of their AI teammates, which ultimately will affect trust at the team level [23]. In fact, recent studies have shown that loss of trust in the AI teammate can affect humans' trust in their human teammate [15, 16], suggesting a vicious cycle of trust loss within a team and organization may be induced by a lack of trust in an AI teammate.

While research has identified a few factors that impact trust in human-AI teams, such as anthropomorphism [4, 21], two limitations exist. First, having been conducted almost exclusively in simulation and laboratory settings, the unique context of the workplace and its associated organizational culture and dynamics, as well as the perspectives of the actual workforce who will be working closely with AI have not been considered. Second, the way trust was conceptualized in this line of research is heavily weighted towards human trust in technology rather than trusting an AI as a colleague. The current study aims to fill this gap by stepping out of research labs to look at the unique context of the work environment. After all, it is the workplace professionals who will be working alongside and collaborate with AI in the real world [24]. Therefore, we pose the following research question: **What dimensions of anthropomorphism are specifically associated with work professionals' perceptions of an AI as a trustworthy colleague?** To answer this question we conducted twenty-two semi-structured interviews with individuals representing a diverse array of work roles within the IT sector.

In this paper, we show that a sense of presence reflective of coworkers, the ability to engage in feedback loop and team processes through human communication, and the ability to self-improve are key components of a trustworthy AI colleague (see Fig.1). We propose suggestions for designing a trustworthy AI colleague into the workplace based on the findings. This research contributes to CSCW by exploring and identifying the unique dimensions of anthropomorphism associated with workplace professionals' trust in AI as a colleague.

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2 METHODS

This study's participants were recruited following their participation in a related study on adaptive autonomy in team work environments based upon two criteria: previous experience with AI technology and diversity of work roles. A total of twenty-two participants were interviewed for this study (5 women, 17 men). Three identified as a non-white ethnic minority, and the rest are Caucasians. Their professions included cyber operators, software engineers, electrical engineers, project managers, and graduate students. Participants were asked to compare and contrast the AI they have interacted with to an ideal AI who they could imagine "working with" and they could trust under various working scenarios. They were also asked to elaborate on the features that make the AI more like a teammate and colleague rather than a tool.

All the interviews were conducted over video conferencing platforms and recorded with the participant's permission. During the interview, the first author followed a semi-structured interview script that allowed for flexible lines of inquiry. Interviews ranged from 15-40 minutes depending on the degree of engagement of the participant. Interviews 20-25 minutes were most common. During the open coding stage, the authors exhausted the data and developed an initial set of codes following the grounded theory approach [22]. The authors then engaged in axial coding [12, 19] and iterated between examining quotes in their context, grouping similar codes to form higher level themes, and uncovering the connections among the themes. These iterations allowed us to identify three main themes that jointly provide a preliminary answer to our research question. These themes are visually summarized in Figure 1.

3 FINDINGS

3.1 AI colleagues need to possess some form of visual presence.

Participants exhibited a uniform desire to "see" the AI. Some explicitly expressed that the AI possessing visual anthropomorphic qualities is a premise for humans to trust it and consider it a colleague and a human counterpart. *"Anthropomorphizing computer technology in general makes them more relate-able, and I think that makes it easier to swallow for most people"* (P18, Male, Network Engineer).

Visual presence can take many forms depending on the contexts and modalities in which human colleagues need to interact with the AI. For instance, P23 believed that it would at least need to be represented using an avatar to be considered a teammate. *"I think to view it as a teammate there might need to be some user interface stuff, where you'd anthropomorphize it"* (P23, Female, Cyber Operations). For some, such virtual representations of the AI would not be enough. These people cared less about anthropomorphism but emphasized the physical form. One participant explained that just his coworkers showing up each day generated a base level of trust. At the very least, *"You trust them to be where they are supposed to be"* (P25, Male, Cyber Operations). P8 echoed this sentiment in his referral to the system as a box somewhere. For him, just having a physical presence would make AI feel like something you could rely upon. *"...think of them as teammates, I mean it'll still be a box*

somewhere but I think they'll become pretty dependent on that" (P8, Male, Cyber Operations).

Taken together, a visual representation seems to be a premise for professionals to view AI as their colleague. This visual representation provides two features that may be equally important for trust development in the AI colleague: human-likeness and physical presence. Human-likeness can be represented virtually in the form of an avatar, and physical presence need not be in human forms. While human desire and preference for anthropomorphism of AI is not new to our current understanding of trust in automation [4, 5], our data suggested that human-likeness may not be the only feature for inducing trust. In fact, physical presence of AI in the form of an object such as a box that is nowhere near human-like can provide humans with a sense of reliability and dependability. So far as it's visible and tangible, and that its physical presence is consistent, it can reduce uncertainty felt by humans [7] and in turn promote trust in the agent.

3.2 AI colleagues need to participate in the feedback loop and social team processes.

Another aspect of trust in AI as a colleague is the AI's ability to give and receive feedback and participate in the team's social processes using human-like communication. Many participants reported that most of the AI technology they had interacted with in their workplace either provided no feedback or the feedback was largely in raw data form. This has made the human-AI relationship felt more user-tool in nature and hindered the formation of trust. For instance, P1 emphasized that in order to consider AI a colleague, it must be capable of giving feedback beyond data form. *"I'd want it to give me feedback in an understandable format, not just give me data. Any tool can give me data"* (P1, Male, Graduate Research Assistant). Many participants indicated a preference for feedback to be in human language. *"Something more interactive will probably help, where it's not just like a program, where you're like this is just data, but something more interesting, words or talking"* (P5, Female, Network Security).

Some participants shared similar sentiments and reported that the feedback to a coworker in the workplace needs to be directly actionable, as opposed to something that requires decoding or decrypting. *"I think I just highlighted that feedback mechanisms probably need to be really realistic and tangible"* (P24, Male, Cyber Operations). He emphasized that the feedback mechanisms would need to replicate that between human colleagues who give one another "constructive feedback", such that the feedback can be incorporated into actions and performance in an efficient manner. Nevertheless, the current AI technologies are almost uniformly programmed to perform one particular task and thus have no awareness of other colleagues' tasks and situations as to provide tangible feedback.

For some, their expectation of an AI colleague to be capable of human communication means more than just the information and data exchanged in human language. Rather, human communication entails shared context or common ground [3]. However, the AI our participants have experienced is not equipped to draw inferences from shared context as humans do, as there is no shared context to begin with. These limitations appeared to have prevented participants from trusting AI the way they trust a human colleague. *"And*

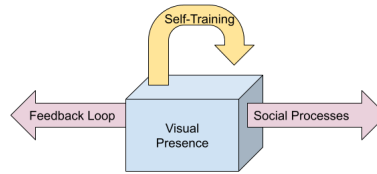


Figure 1: Components of an AI as a trustworthy colleague

a lot of times I do feel like AI is missing that human context element that would be important to have" (P17, Female, Graduate Research Assistant).

More importantly, such human communication is the key means through which team members and coworkers engage in social and interpersonal processes that allow them to develop a sense of team and collaboration [14]. It is through these social processes that trust among team members can be fostered [13]. Their trust in AI can only rely on the evaluation of its ability, which very much resembles trust in automation as a tool [9] rather than as a colleague. *"Lacks the humanity some people associate with teaming and that would be the more difficult aspect"* (P4, Male, Graduate Research Assistant). One participant explicitly cited a team formation theory of human-human teams, regretfully stating that such unique human processes might not be applicable for AI colleagues. *"Like the process for team building dynamics, you start seeing some more of that, the storming, forming, norming"* (P20, Male, Government Contractor).

In sum, professionals in the workplace develop trust in their team members not only based on their technical skills and abilities, but also based on their perception of the colleague's benevolence and integrity [9], generated gradually through social and interpersonal processes achieved through human communication. Much of this human communication involves feedback in terms of evaluative information about one another's performance for improvement purposes [18]. This feedback loop is a vital part of a professional relationship.

3.3 AI colleagues need to be capable of self-improvement.

A final theme, related to the performance improvement mentioned above, is the need for AI teammates to self-improve and self-train without human supervision. After all, in the workplace professionals are expected to grow in their abilities and skills relevant to their job. For instance, P2 below speaks about how tools are just things that come and go as technology improves and the needs of the organization change. Colleagues are able to leverage lessons learned through experience in order to become more value added to the team and, perhaps, even irreplaceable. *"I think it would need to change over time. Tools we replace as they become outdated, so it would need to overcome that and constantly improve"* (P2, Female, IT Project Management).

Similarly, P14 expressed that he would "trust an AI as someone who works alongside" him if its system could understand where it needed to improve and train itself to do that without being directed to do so. *"To train itself, or make itself better, I think that would be a good case here"* (P14, Male, Cyber Operations). Self-training,

he explained, is something that he expects from someone in the professional workplace that he might not expect in other situations, such as in a recreational team.

Other participants made similar statements about the AI needing to be able to teach itself, and one discussed it in terms of evolution. *"In a team everyone really needs to be able to learn and evolve. That's what makes someone a professional, is that desire to grow and improve"* (P6, Female, Computer Systems Engineering). In this quote, P6 explains that part of being in a professional team is that you are expected to seek self-improvement and develop your skills. As a technical professional, she thought it would be very feasible to have a system self-evaluate its work for flaws and train to address those flaws. Without these capabilities, she could not trust it not to fall behind and repeatedly make the same mistake. She added that in the workplace, *"that kind of (human) coworker would be fired"*.

4 DESIGN RECOMMENDATIONS

Enhancing the presence of AI colleagues. Our data suggests that it is hard for professionals to trust an AI colleague whom they cannot see. The visual representation of an AI colleague should reflect that of the human workers within the work context, at least to an extent (e.g. a human-like virtual avatar, or an object that "shows up to work"), to give a sense of presence. Therefore, to promote trust in AI as a colleague in the workplace, we propose that the system should support and enhance any type of presence outlined in computer-mediated communication literature (e.g., a sense of shared space, realism, social presence, etc.[11]). For instance, an attendance-taking feature can be implemented for professionals to keep track, where an AI colleague is expected to be "in" every workday but is also allowed to "take days off" for maintenance.

AI colleague engaged in team building. Second, professionals desired team processes achieved through human communication and actionable feedback from their colleagues in order to trust them. With the advancement of natural language processing and beyond, an AI colleague should be equipped with team building skills to follow if not lead different stages of team formation and development, as to be considered and trusted as a real colleague.

AI colleague demonstrating growth over time. Last but not least, it is expected that in the workplace, professionals should grow in their roles and self-improve, and so should an AI colleague. We propose that an AI teammate should be designed to iteratively- either at the completion of a work cycle, or at the completion of set time intervals- evaluate its performance for incompetency and capability gaps and train to fill them. This can be done through consulting available resources, organizational repository of training materials, etc.. Then it should re-evaluate its skills against task objectives

and the organization's goals for areas of improvement. These self-training abilities and efforts should be made known to its human colleagues. One way to do this would be to have the AI submit the same period progress reports that its human colleagues do, but make the AI's reports visible to the entire work team.

5 LIMITATIONS AND FUTURE WORK

A limitation of this study is the participant population, which is largely composed of technical professionals due to the prior study from which they were selected. It is important that future work also consider the perceptions of less technical individuals who would possibly be more uncomfortable with AI technology in general, as research shows that people unfamiliar with a new technology require substantial trust in the technology's ability to perform its tasks in order to accept it [8].

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REFERENCES

- [1] Tammy Bahmanziari, J Michael Pearson, and Leon Crosby. 2003. Is trust important in technology adoption? A policy capturing approach. *Journal of Computer Information Systems* 43, 4 (2003), 46–54.
- [2] Gagan Bansal, Tongshuang Wu, Joyce Zhou, Raymond Fok, Besmira Nushi, Ece Kamar, Marco Tulio Ribeiro, and Daniel Weld. 2021. Does the whole exceed its parts? the effect of ai explanations on complementary team performance. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [3] Herbert H Clark and Susan E Brennan. 1991. Grounding in communication. (1991).
- [4] Myke C. Cohen, Mustafa Demir, Erin K. Chiou, and Nancy J. Cooke. 2021. The Dynamics of Trust and Verbal Anthropomorphism in Human-Autonomy Teaming. In *2021 IEEE 2nd International Conference on Human-Machine Systems (ICHMS)*. IEEE, Magdeburg, Germany, 1–6. <https://doi.org/10.1109/ICHMS53169.2021.9582655>
- [5] Ewart J. de Visser, Samuel S. Monfort, Ryan McKendrick, Melissa A. B. Smith, Patrick E. McKnight, Frank Krueger, and Raja Parasuraman. 2016. Almost human: Anthropomorphism increases trust resilience in cognitive agents. *Journal of Experimental Psychology: Applied* 22, 3 (Sept. 2016), 331–349. <https://doi.org/10.1037/xap0000092>
- [6] Alpana Dubey, Kumar Abhinav, Sakshi Jain, Veenu Arora, and Asha Puttaveerana. 2020. HACO: a framework for developing human-AI teaming. In *Proceedings of the 13th Innovations in Software Engineering Conference on Formerly known as India Software Engineering Conference*. 1–9.
- [7] Mary T Dzindolet, Scott A Peterson, Regina A Pomranky, Linda G Pierce, and Hall P Beck. 2003. The role of trust in automation reliance. *International journal of human-computer studies* 58, 6 (2003), 697–718.
- [8] Chang Lee and Guohua Wan. 2010. Including subjective norm and technology trust in the technology acceptance model: a case of e-ticketing in China. *ACM SIGMIS Database: The DATABASE for Advances in Information Systems* 41, 4 (2010), 40–51.
- [9] John D Lee and Katrina A See. 2004. Trust in automation: Designing for appropriate reliance. *Human factors* 46, 1 (2004), 50–80.
- [10] Brian Y Lim, Anind K Dey, and Daniel Avrahami. 2009. Why and why not explanations improve the intelligibility of context-aware intelligent systems. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 2119–2128.
- [11] Matthew Lombard and Theresa Ditton. 1997. At the Heart of It All: The Concept of Presence. *Journal of Computer-Mediated Communication* 3, 2 (Sept. 1997). <https://doi.org/10.1111/j.1083-6101.1997.tb00072.x>
- [12] Joseph A. Maxwell. 2012. *Qualitative research design: An interactive approach*. Vol. 41. SAGE Publications, Inc.
- [13] Roger C Mayer, James H Davis, and F David Schoorman. 1995. An integrative model of organizational trust. *Academy of management review* 20, 3 (1995), 709–734.
- [14] Daniel McAllister, J. 1995. Affect- and cognition-based trust as foundations for interpersonal cooperation in organizations. *The Academy of Management Review* 38, 1 (1995), 24–59.
- [15] Nathan J McNeese, Mustafa Demir, Erin Chiou, Nancy Cooke, and Giovanni Yanikian. 2019. Understanding the Role of Trust in Human-Autonomy Teaming. In *Proceedings of the 52nd Hawaii International Conference on System Sciences*. 254–263.
- [16] Nathan J. McNeese, Mustafa Demir, Erin K. Chiou, and Nancy J. Cooke. 2021. Trust and Team Performance in Human-Autonomy Teaming. *International Journal of Electronic Commerce* 25, 1 (Jan. 2021), 51–72. <https://doi.org/10.1080/10864415.2021.1846854>
- [17] Raja Parasuraman and Victor Riley. 1997. Humans and automation: Use, misuse, disuse, abuse. *Human factors* 39, 2 (1997), 230–253.
- [18] Habiba Saedon, Shizalia Salleh, Arun Balakrishnan, Christopher HE Imray, and Mahmud Saedon. 2012. The role of feedback in improving the effectiveness of workplace based assessments: a systematic review. *BMC medical education* 12, 1 (2012), 1–8.
- [19] Cliff Scott and Melissa Medaugh. 2017. Axial coding. *The international encyclopedia of communication research methods* 10 (2017), 9781118901731.
- [20] Tenace Kwaku Setor. 2014. The effect of job stress on job performance amongst it professionals: The moderating role of proactive work behaviours. In *Proceedings of the 52nd ACM conference on Computers and people research*. 17–21.
- [21] William Seymour and Max Van Kleek. 2021. Exploring Interactions Between Trust, Anthropomorphism, and Relationship Development in Voice Assistants. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 371 (oct 2021), 16 pages. <https://doi.org/10.1145/3479515>
- [22] Anselm Strauss and Juliet Corbin. 1994. Grounded theory methodology: An overview. (1994).
- [23] Anna-Sophie Ulfert and Eleni Georganta. 2020. A model of team trust in human-agent teams. In *Companion publication of the 2020 international conference on multimodal interaction*. 171–176.
- [24] Dakuo Wang, Elizabeth Churchill, Pattie Maes, Xiangmin Fan, Ben Shneiderman, Yuanchun Shi, and Qianying Wang. 2020. From Human-Human Collaboration to Human-AI Collaboration: Designing AI Systems That Can Work Together with People. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3334480.3381069>
- [25] Fumeng Yang, Zhuanyi Huang, Jean Scholtz, and Dustin L Arendt. 2020. How do visual explanations foster end users' appropriate trust in machine learning?. In *Proceedings of the 25th International Conference on Intelligent User Interfaces*. 189–201.
- [26] Rui Zhang, Nathan J McNeese, Guo Freeman, and Geoff Musick. 2021. "An Ideal Human" Expectations of AI Teammates in Human-AI Teaming. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW3 (2021), 1–25.